MINIMUM VIABLE PRODUCT (MVP) & BUSINESS REQUIREMENTS DOCUMENT (BRD)

Agentic AI Insights Platform

Business & Technical Overview for Corporate Banking Use Cases

1. Executive Summary

The **Agentic AI Insights Platform** is a lightweight decision-support tool designed to help business and risk stakeholders quickly extract meaningful insights from operational and customer data. Built using Python, SQL, Mistral, and Flask, the platform combines structured analytics with agentic reasoning to generate clear, auditable summaries, trends, and risk indicators.

The purpose of the platform is to demonstrate how AI-enabled analysis can support governance, risk, and control activities within a corporate banking environment. It provides a foundation for responsible, explainable insight generation while maintaining alignment with regulatory expectations and business value. The tool is intentionally scoped as an MVP to showcase practical technical capability alongside strong business understanding — reflecting the hybrid skillset required in roles such as HRBC, PM, and business-aligned analytics functions within HSBC.

2. MVP Definition

Problem Statement

> *Corporate and institutional banking teams often rely on manual analysis, fragmented data sources, and inconsistent reporting to understand customer behaviour, emerging risks, and operational trends. This creates delays, increases operational risk, and makes it harder for stakeholders to make timely, informed decisions. There is a need for a simple, auditable tool that can surface insights quickly while supporting governance and control expectations.*

Target Users

- **Risk & Control Analysts:** Seeking early indicators of anomalies or emerging issues.
- **Business Stakeholders / Relationship Managers:** Needing quick customer or portfolio insights.
- **Project Managers / HRBCs:** Who require structured, explainable analysis to support decision-making.

MVP Scope

The MVP delivers a minimal but functional version of the platform with the following capabilities:

CAPABILITY AREA	DESCRIPTION
Agentic Workflow	Natural-language querying of structured data via an agentic AI workflow.
Data Retrieval	SQL-driven retrieval of customer or operational insights.
Insight Generation	Clear, concise summaries of trends, patterns, and anomalies.
Risk Scoring	Basic risk-focused outputs (e.g., flags, simple scoring logic).
User Interface	A lightweight Flask interface for interaction.
Auditability	Transparent reasoning steps to support governance and auditability.

Success Metrics

- Ability to answer core business questions consistently and accurately.
- Reduction in manual effort required to generate basic insights.
- Clear alignment to governance, risk, and control expectations.
- Positive feedback from stakeholders on clarity and usefulness of outputs.
- Demonstration of technical competency in Python, SQL, agentic AI, and application design.

3. Business Requirements

3.1 Functional Requirements

ID	REQUIREMENT NAME	DESCRIPTION
FR1	Natural Language Querying	The system must allow users to submit natural-language questions and convert them into structured analytical tasks using an agentic workflow.
FR2	Data Retrieval via SQL	The system must retrieve relevant data from the underlying SQL database based on the interpreted query.
FR3	Insight Generation	The system must produce concise, business-focused insights, including trends, summaries, anomalies, and simple risk indicators.
FR4	Agentic Reasoning Chain	The system must provide a transparent reasoning process (e.g., chain-of-thought summary or structured explanation) to support governance and auditability.
FR5	Flask-Based User Interface	The system must provide a lightweight web interface for submitting queries and viewing results.
FR6	Error Handling & Safe Responses	The system must detect unsupported or ambiguous queries and return safe, controlled responses rather than speculative or non-compliant outputs.
FR7	Extensibility for Risk Features	The system must support the addition of future risk-related capabilities such as anomaly detection, risk scoring, or governance guardrails.

3.2 Non-Functional Requirements

🔒 **SECURITY & GOVERNANCE GOVERNANCE HIGHLIGHTS**

- **NFR1 — Transparency & Explainability:** Outputs must include a clear explanation of how insights were generated to support governance, risk and control expectations.
- **NFR5 — Security & Data Handling:** The system must avoid exposing sensitive data and follow principles of data minimisation. No personally identifiable information (PII) should be stored unnecessarily.

- **NFR2 — Consistency:** The system must produce consistent insights when given similar inputs, reducing variability in manual analysis.

- **NFR3 — Auditability:** The system must be capable of logging queries, data sources accessed, and generated outputs for audit and compliance purposes (future enhancement).
- **NFR4 — Performance:** Responses should be generated within a reasonable timeframe (e.g., under 5 seconds for standard queries).
- **NFR6 — Reliability:** The system should handle unexpected inputs gracefully and avoid system crashes or unhandled exceptions.
- **NFR7 — Maintainability:** The codebase should be modular and documented to support future enhancements, including risk-focused features and domain-specific logic.

3.3 Constraints

- The MVP uses synthetic or non-sensitive data to avoid regulatory issues.
- The system is not intended to replace human judgement; it supports decision-making.
- The MVP is scoped for demonstration and learning purposes, not production deployment.
- Cloud deployment is optional at this stage and may be limited by available resources.

3.4 Assumptions

- Users have basic familiarity with business insights and risk concepts.
- Data provided to the system is clean, structured, and suitable for SQL-based analysis.
- Future enhancements may require integration with additional data sources or APIs.
- Governance, risk and control requirements will evolve as the tool matures.

4. Governance, Risk, Controls & HSBC Themes

4.1 Governance

Effective governance ensures that analytical and AI-enabled tools operate in a transparent, consistent and accountable manner. The platform is designed to support governance expectations by providing clear, structured explanations of how insights are generated, ensuring consistent logic and repeatable outputs for similar queries, and enabling traceability of data sources and reasoning steps. This oversight makes analytical processes visible rather than opaque, reducing ambiguity and fostering organizational trust.

4.2 Risk

Risk management is central to any environment where decisions impact customers, financial exposure or operational stability. The platform contributes to risk awareness and mitigation by highlighting anomalies, unusual patterns, or emerging trends in data, supporting early identification of potential portfolio issues. It reduces manual interpretation risk through standardized analysis and structured summaries, minimizing the probability of oversight or human error.

4.3 Controls

Controls ensure that analytical tools operate safely, reliably and within defined boundaries. The platform incorporates control-aligned principles such as query-rejection guardrails, fallback safe responses for ambiguous inputs, strict data minimization, and clear structural separation between data retrieval (SQL), reasoning (LLM), and presentation layers.

4.4 Corporate Banking Use Cases

- **Customer insight summaries** for relationship managers to execute quick portfolio health reviews.
- **Early warning indicators** for portfolio health, flagging emerging anomalies.
- **Operational risk pattern detection** across operational layers to track system behavior.
- **Trend analysis** across specific commercial customer segments or product categories.

5. Planned Enhancements (Risk-Focused Features)

The next phase of development focuses on strengthening the platform's ability to support risk-aware decision-making, improve analytical depth, and enhance governance and control alignment. These enhancements reflect the needs of regulated, data-driven environments.

- **5.1 Risk Scoring Framework:** Introduce a transparent risk scoring mechanism that evaluates data against predefined rules or weighted indicators, detailing exactly how each score was derived.
- **5.2 Anomaly Detection:** Add basic anomaly detection logic (such as statistical deviation checks like z-scores or moving averages) to isolate outliers or unexpected pattern breaks.
- **5.3 Governance Guardrails:** Enhance safe-response mechanics to reject out-of-scope or speculative queries, restricting access to sensitive data and enforcing baseline uncertainty metrics.
- **5.4 Audit Logging and Traceability:** Introduce lightweight logging to systematically record user queries, exact SQL code executed, timestamps, and model reasoning traces.
- **5.5 Explainability Enhancements:** Provide feature attribution highlights to detail exactly which metrics influenced outputs, alongside standard confidence indicators.
- **5.6 Expanded Analytical Features:** Support multi-period trend analysis, cohort benchmarking, peer-group summaries, and basic what-if modeling.
- **5.7 Integration with Business Tools:** Develop structured JSON/CSV data exports formatted for direct ingestion by enterprise business intelligence suites like **Power BI**.

6. Technical Foundations

The platform is built on a lightweight, modular technical stack designed to demonstrate practical capability in AI-enabled analysis, data handling, and application development.

6.1 Core Technologies

TECHNOLOGY	ROLE IN PLATFORM	KEY FUNCTIONS
Python	Core Orchestration & Logic	Data manipulation, processing, orchestration logic, rule execution, and LLM model integration.
SQL	Structured Data Retrieval	Querying customer/operational datasets, aggregations, and maintaining a verifiable data retrieval layer.
Mistral (LLM)	Agentic Reasoning & NLU	Interpreting user intent, compiling structured analytical tasks, and outputting reasoning traces.
Flask	Web Interface & Routing	Providing a simple web front end for input submission, rendering analytical results and logic paths.

6.2 Supporting Technologies (Planned Roadmap)

- **Cloud Basics (Planned):** Cloud fundamentals (e.g., Azure/AWS) will support future containerised deployment, scalable hosting, and enterprise-grade data access controls.
- **Power BI (Planned):** Used to bridge the gap between AI-generated insights and stakeholder presentation, enabling dynamic charts, risk dials, and automated reporting dashboards.

6.3 Design Principles

The design framework follows five core principles:

Transparency	Modularity	Auditability	Safety
Business & Governance Alignment			